

ULJIN, Korea, Republic of

WMO: 471300

Lat: 36.9917N

Lon: 129.4128E

Elev: 168

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	22.4	-9.9	3.3	23.7	-5.9	4.0	26.0	22.8	34.7	20.0	35.4	8.9	230	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	87.6	75.6	84.6	74.7	82.4	73.9	78.2	83.8	76.9	82.5	75.6	80.9	7.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	139.9	81.3	75.2	133.0	80.2	73.8	126.9	79.2	41.9	83.7	40.6	82.6	39.3	81.1	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	19.6	16.7	14.6	13.9	93.5	4.1	2.3	10.9	95.2	8.5	96.6	6.2	97.9	3.2	99.6	
(5)				10.1	79.4	3.7	1.9	7.4	80.8	5.3	81.9	3.2	82.9	0.5	84.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	34.8	37.4	44.3	53.6	61.3	66.9	74.1	75.6	67.8	59.4	49.0	38.9	(6)
(7)		DBStd	14.97	5.87	6.32	6.70	6.98	5.97	4.77	5.55	4.67	4.27	5.07	6.44	6.51	(7)
(8)		HDD50	1514	471	356	204	36	0	0	0	0	0	4	96	347	(8)
(9)		HDD65	4370	936	774	643	348	147	31	1	0	18	182	481	809	(9)
(10)		CDD50	3471	0	2	27	145	352	506	747	793	535	296	65	3	(10)
(11)		CDD65	851	0	0	0	8	34	87	283	328	102	9	0	0	(11)
(12)		CDH74	4958	0	0	3	46	173	293	1870	2228	333	12	0	0	(12)
(13)		CDH80	1126	0	0	0	9	29	48	502	500	38	0	0	0	(13)
(14)	Wind	WSAvg	7.1	8.2	7.9	8.0	8.0	6.9	5.9	6.0	5.9	6.7	7.0	7.5	7.8	(14)
(15)	Precipitation	PrecAvg	43.7	2.0	2.3	2.8	3.1	2.5	3.9	6.1	7.8	6.2	3.2	2.4	1.7	(15)
(16)		PrecMax	61.1	5.6	7.1	6.6	7.3	5.9	8.0	12.8	20.2	13.7	9.1	7.5	7.9	(16)
(17)		PrecMin	26.9	0.0	0.2	0.2	0.9	0.4	0.4	1.5	0.9	0.4	0.2	0.1	0.0	(17)
(18)		PrecStd	9.6	1.7	1.8	1.6	1.7	1.6	2.0	3.2	4.9	3.7	2.4	2.1	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	58.4	69.6	79.3	84.0	85.5	91.9	92.6	84.4	75.4	69.0	59.6	(19)
(20)			MCWB	43.5	46.6	52.3	58.0	62.1	67.3	76.4	77.2	74.0	63.6	56.3	49.6	(20)
(21)		2%	DB	49.3	54.2	62.9	73.1	78.6	80.2	88.0	87.6	80.0	72.3	65.4	54.9	(21)
(22)			MCWB	40.1	44.8	49.3	55.8	60.0	67.4	75.6	77.2	72.0	63.0	55.4	45.8	(22)
(23)		5%	DB	46.5	51.0	58.9	68.4	74.9	77.1	85.0	84.7	77.5	70.2	62.5	51.9	(23)
(24)			MCWB	38.3	42.3	48.1	53.8	59.1	67.1	75.2	76.2	70.5	61.9	53.8	43.3	(24)
(25)		10%	DB	44.0	47.8	55.4	64.4	71.4	74.5	82.6	82.5	75.4	68.4	59.7	49.2	(25)
(26)			MCWB	36.7	40.1	46.3	52.4	58.7	66.6	74.6	75.4	69.1	60.9	51.6	41.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	51.5	55.8	62.5	66.3	72.7	79.3	80.6	76.3	69.1	60.9	52.2	(27)
(28)			MCD B	49.4	54.4	63.7	73.0	74.1	78.2	85.4	85.9	80.5	72.1	64.3	57.1	(28)
(29)		2%	WB	42.4	46.7	52.4	58.9	63.7	70.5	77.8	78.8	74.0	66.1	57.7	47.7	(29)
(30)			MCD B	46.8	51.9	59.7	67.3	72.0	75.7	84.3	84.3	78.7	69.5	62.6	53.3	(30)
(31)		5%	WB	40.0	43.5	49.7	56.6	62.1	69.2	76.5	77.5	71.9	64.0	55.3	44.8	(31)
(32)			MCD B	44.6	49.4	56.1	64.5	69.5	74.1	83.2	82.7	76.0	68.3	60.9	50.2	(32)
(33)		10%	WB	37.7	40.9	47.3	54.8	60.9	67.9	75.2	76.4	70.2	62.1	52.8	42.2	(33)
(34)			MCD B	42.7	46.6	54.3	61.3	67.6	72.3	81.6	81.3	74.1	67.0	58.2	47.9	(34)
(35)	Mean Daily Temperature Range	MDBR	14.2	14.2	14.7	14.7	13.2	9.9	8.9	9.2	11.7	14.8	15.3	14.9	(35)	
(36)		5% DB	MCD BR	15.9	17.4	19.7	20.1	19.4	14.4	13.5	12.5	13.8	16.1	17.3	17.0	(36)
(37)			MCWBR	11.3	11.9	12.4	11.0	9.4	6.9	5.8	6.2	8.3	10.2	11.7	11.9	(37)
(38)		5% WB	MCD BR	13.5	15.6	16.3	16.1	14.7	11.0	11.8	10.5	11.8	12.8	14.6	15.2	(38)
(39)			MCWBR	11.1	12.2	12.3	10.8	8.9	6.2	6.1	6.2	7.9	9.5	11.2	12.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.423	0.485	0.580	0.613	0.620	0.662	0.625	0.567	0.497	0.482	0.447	0.413	(40)	
(41)		taud	1.994	1.874	1.696	1.681	1.708	1.685	1.830	1.969	2.097	2.006	2.019	2.035	(41)	
(42)		Ebn at Noon	228	228	218	221	223	213	220	230	240	227	218	221	(42)	
(43)		Edh at Noon	43	54	71	76	75	77	66	56	47	47	41	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	785	1006	1309	1550	1710	1603	1402	1381	1205	1066	790	710	(44)	
(45)		RadStd	63	112	124	112	166	138	195	161	139	103	104	59	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (27 neighbors)	N/A	N/A	-1.85	N/A	N/A	N/A	N/A	+166	+88	(47)

Nomenclature: See separate page